

ForgeCloud AWS, Terraform & Networking

Cloud architecture, Infrastructure as Code and request flow

1. AWS Architecture

AWS is the primary cloud platform. Terraform automates the creation and update of VPC, IAM, EKS, load balancing and supporting resources.

2. Terraform Modules

```
terraform/  
  modules/  
    vpc/  
    networking/  
    iam/  
    eks/  
    ecr/  
  
  environments/  
    dev/  
    prod/  
  
main.tf  
variables.tf  
outputs.tf  
providers.tf
```

3. AWS Resource Flow

```
Terraform  
|  
+--> VPC  
|   +--> Public Subnets  
|   +--> Private Subnets  
|   +--> Route Tables  
|   +--> Internet/NAT  
|  
+--> IAM Roles  
|  
+--> EKS Cluster  
|   +--> Node Groups  
|  
+--> ECR  
|  
+--> Load Balancer / supporting networking
```

4. Network Request Path

```
Client  
|  
DNS  
|  
Load Balancer  
|  
Ingress  
|  
Kubernetes Service  
|  
Application Pod  
|  
Database
```

5. Networking Concepts

- TCP/IP communication.
- HTTP/HTTPS and TLS.
- DNS resolution.
- Ports.
- VPC architecture.
- Public and private subnets.
- Routing and route tables.
- Security groups.
- Load balancing.
- Ingress and service discovery.
- Kubernetes network policies.

6. Terraform Safety

- Run terraform fmt.
- Run terraform validate.
- Review terraform plan before apply.
- Never hard-code AWS credentials.
- Use least-privilege IAM.
- Do not run terraform destroy automatically.
- Document cleanup steps and potentially expensive resources.